

E1 Trigonometry — Answer Key

IGCSE Mathematics 0580 | Full Solutions

E1_Q1: Pythagoras Theorem - Problems

F Foundation — Answer: 5 cm

Using Pythagoras theorem: $c^2 = a^2 + b^2 = 3^2 + 4^2 = 9 + 16 = 25$. So $c = \sqrt{25} = 5$ cm.

F Foundation — Answer: 12 cm

Using Pythagoras theorem: $a^2 = c^2 - b^2 = 13^2 - 5^2 = 169 - 25 = 144$. So $a = \sqrt{144} = 12$ cm.

F Foundation — Answer: 10 cm

$c^2 = 6^2 + 8^2 = 36 + 64 = 100$. $c = \sqrt{100} = 10$ cm.

F Foundation — Answer: 15 cm

$c^2 = 9^2 + 12^2 = 81 + 144 = 225$. $c = \sqrt{225} = 15$ cm. This is a 3-4-5 triple scaled by 3.

C Core — Answer: 4.8 m

Let h be the height up the wall. $h^2 = 5^2 - 1.4^2 = 25 - 1.96 = 23.04$. $h = \sqrt{23.04} = 4.8$ m.

C Core — Answer: 100 m

The diagonal forms the hypotenuse of a right triangle with legs 80 m and 60 m. $d^2 = 80^2 + 60^2 = 6400 + 3600 = 10000$. $d = \sqrt{10000} = 100$ m.

C Core — Answer: 26 km

The distance forms the hypotenuse of a right triangle with legs 24 km and 10 km. $d^2 = 24^2 + 10^2 = 576 + 100 = 676$. $d = \sqrt{676} = 26$ km.

C Core — Answer: 13 cm

The 3D diagonal is given by $d^2 = l^2 + w^2 + h^2 = 3^2 + 4^2 + 12^2 = 9 + 16 + 144 = 169$. $d = \sqrt{169} = 13$ cm.

C Core — Answer: $\sqrt{84} = 2\sqrt{21}$ cm or approximately 9.17 cm

Half of the base side is 4 cm. The slant height, half-base, and pyramid height form a right triangle. $h^2 = 10^2 - 4^2 = 100 - 16 = 84$. $h = \sqrt{84} = 2\sqrt{21}$ cm.

C Core — Answer: $\sqrt{23400} = 30\sqrt{26}$ m or approximately 153.0 m

The horizontal distance between A and B is 150 m. The vertical difference in height is $70 - 40 = 30$ m. The straight-line distance between C and D: $d^2 = 150^2 + 30^2 = 22500 + 900 = 23400$. $d = \sqrt{23400} = 30\sqrt{26}$ m.

C Core — Answer: $\sqrt{70}$ m or approximately 8.37 m

The longest rod fits along the 3D diagonal. $d^2 = 6^2 + 5^2 + 3^2 = 36 + 25 + 9 = 70$. $d = \sqrt{70}$ m.

C Core — Answer: 30 feet

Using Pythagoras: $d^2 = 50^2 - 40^2 = 2500 - 1600 = 900$. $d = \sqrt{900} = 30$ feet.

H Challenge — Answer: $x = 3$

The 3D diagonal: $d^2 = (2x)^2 + (3x)^2 + (6x)^2 = 4x^2 + 9x^2 + 36x^2 = 49x^2$. So $d = 7x$. Given $d = 21$, so $7x = 21$, $x = 3$.

H Challenge — Answer: $x = \sqrt{10}$

By Pythagoras: $(x + 1)^2 + (x + 5)^2 = (x + 6)^2$. Expand: $(x^2 + 2x + 1) + (x^2 + 10x + 25) = x^2 + 12x + 36$. Simplify: $2x^2 + 12x + 26 = x^2 + 12x + 36$. Subtract: $x^2 = 10$. So $x = \sqrt{10}$.

H Challenge — Answer: $\sqrt{73}$ km or approximately 8.54 km

Net east displacement: $12 - 8 = 4$ km east. Net north-south displacement: 5 km south. Distance PS = $\sqrt{4^2 + 5^2} = \sqrt{16 + 25} = \sqrt{73}$ km.

H Challenge — Answer: $\sqrt{100/3} = (10\sqrt{3})/3$ cm or approximately 5.77 cm

Space diagonal $d = \sqrt{8^2 + 6^2 + h^2} = \sqrt{64 + 36 + h^2} = \sqrt{100 + h^2}$. Given $d = 2h$, so $2h = \sqrt{100 + h^2}$. Square both sides: $4h^2 = 100 + h^2 \Rightarrow 3h^2 = 100 \Rightarrow h = \sqrt{100/3} = 10/\sqrt{3} = (10\sqrt{3})/3$ cm.

E1_Q2: Trigonometric Ratios - Problems

F Foundation — Answer: $3/5$ or 0.6

$\sin(\theta) = \text{opposite/hypotenuse} = 3/5 = 0.6$.

F Foundation — Answer: $5/13$ or approximately 0.3846

$\cos(\theta) = \text{adjacent/hypotenuse} = 5/13$.

F Foundation — Answer: $8/15$ or approximately 0.5333

$\tan(\theta) = \text{opposite/adjacent} = 8/15$.

F Foundation — Answer: $7/25$ or 0.28

$\sin(\theta) = \text{opposite/hypotenuse} = 7/25 = 0.28$.

C Core — Answer: 30 degrees

$\theta = \sin^{-1}(0.5) = 30$ degrees.

C Core — Answer: 53.1 degrees

Angle ACB is at C. $\tan(\text{ACB}) = \text{opposite}/\text{adjacent} = \text{AB}/\text{BC} = 8/6 = 4/3$. angle ACB = $\tan^{-1}(4/3) = 53.13\dots$ degrees. Rounded to 1 dp: 53.1 degrees.

C Core — Answer: 6.43 cm

$\sin(40) = \text{opposite}/\text{hypotenuse} = \text{opposite}/10$. opposite = $10 * \sin(40) = 10 * 0.6428 = 6.428$ cm. Rounded to 2 dp: 6.43 cm.

C Core — Answer: 14.65 cm

$\cos(35) = \text{adjacent}/\text{hypotenuse} = 12/h$. $h = 12 / \cos(35) = 12 / 0.81915 = 14.649$ cm. Rounded to 2 dp: 14.65 cm.

C Core — Answer: 15.01 cm

$\cos(28) = \text{adjacent}/17$. adjacent = $17 * \cos(28) = 17 * 0.88295 = 15.010$ cm. Rounded to 2 dp: 15.01 cm.

C Core — Answer: 36.9 degrees

Angle PRQ is at R. We need to find angle at R. $\sin(\text{PRQ}) = \text{opposite}/\text{hypotenuse} = \text{PQ}/\text{PR} = 9/15 = 3/5 = 0.6$. angle PRQ = $\sin^{-1}(0.6) = 36.87\dots$ degrees. Rounded to 1 dp: 36.9 degrees.

C Core — Answer: 8.83 m

$\sin(65) = \text{opposite}/\text{hypotenuse} = 8/\text{length}$. length = $8 / \sin(65) = 8 / 0.90631 = 8.827$ m. Rounded to 2 dp: 8.83 m.

C Core — Answer: 12.12 m

$\tan(22) = \text{height}/30$. height = $30 * \tan(22) = 30 * 0.40403 = 12.121$ m. Rounded to 2 dp: 12.12 m.

H Challenge — Answer: $\sin A = 4/5$, $\tan A = 4/3$

If $\cos A = 3/5 = \text{adjacent}/\text{hypotenuse}$, then adjacent = $3k$ and hypotenuse = $5k$ for some k . By Pythagoras, $\text{opposite}^2 = (5k)^2 - (3k)^2 = 25k^2 - 9k^2 = 16k^2$, so opposite = $4k$. Therefore $\sin A = \text{opposite}/\text{hypotenuse} = 4k/5k = 4/5$, and $\tan A = \text{opposite}/\text{adjacent} = 4k/3k = 4/3$.

H Challenge — Answer: $\cos \theta = 4/5$, $\tan \theta = 3/4$

$5 \sin \theta = 3$, so $\sin \theta = 3/5 = \text{opposite}/\text{hypotenuse}$. Let opposite = $3k$, hypotenuse = $5k$. By Pythagoras, adjacent = $\sqrt{(5k)^2 - (3k)^2} = \sqrt{16k^2} = 4k$. So $\cos \theta = \text{adjacent}/\text{hypotenuse} = 4k/5k = 4/5$, and $\tan \theta = \text{opposite}/\text{adjacent} = 3k/4k = 3/4$.

H Challenge — Answer: 15.00 cm

Let the legs be a and b , with the angle opposite side a being 36.87 degrees. Then $\tan(36.87) = a/b$, so $a = b * \tan(36.87) = b * 0.75 = 0.75b$. Area = $(1/2) * a * b = (1/2) * 0.75b * b = 0.375b^2 = 54$. So $b^2 = 54/0.375 = 144$, $b = 12$. $a = 0.75 * 12 = 9$. Hypotenuse = $\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.00$ cm.

H Challenge — Answer: $\sin \theta = 3/5$, $\cos \theta = 4/5$

$\tan \theta = \text{opposite}/\text{adjacent} = 3/4$. Let opposite = $3k$ and adjacent = $4k$. Hypotenuse = $\sqrt{(3k)^2 + (4k)^2} = \sqrt{9k^2 + 16k^2} = \sqrt{25k^2} = 5k$. So $\sin \theta = \text{opposite}/\text{hypotenuse} = 3k/5k = 3/5$, and $\cos \theta = \text{adjacent}/\text{hypotenuse} = 4k/5k = 4/5$.

E1_Q3: Angle of Elevation and Depression - Problems

F Foundation — Answer: 11.5 m

$\tan(30) = \text{height}/20$. height = $20 * \tan(30) = 20 * 0.57735 = 11.547$ m. Rounded to 1 dp: 11.5 m.

F Foundation — Answer: 12 m

$\tan(45) = \text{height}/12$. height = $12 * \tan(45) = 12 * 1 = 12$ m.

F Foundation — Answer: 86.6 m

Angle of depression = 30 degrees equals angle of elevation from boat to top. $\tan(30) = 50/\text{distance}$. distance = $50/\tan(30) = 50/0.57735 = 86.603$ m. Rounded to 1 dp: 86.6 m.

F Foundation — Answer: 17.3 m

$\tan(60) = 30/\text{distance}$. distance = $30/\tan(60) = 30/1.73205 = 17.321$ m. Rounded to 1 dp: 17.3 m.

C Core — Answer: 15.62 m

$\tan(32) = \text{height}/25$. height = $25 * \tan(32) = 25 * 0.62487 = 15.622$ m. Rounded to 2 dp: 15.62 m.

C Core — Answer: 184.67 m

Angle of depression = 18 degrees. $\tan(18) = 60/\text{distance}$. distance = $60/\tan(18) = 60/0.32492 = 184.663$ m. Rounded to 2 dp: 184.67 m.

C Core — Answer: 48.83 m

$\sin(55) = 40/\text{string_length}$. string_length = $40/\sin(55) = 40/0.81915 = 48.831$ m. Rounded to 2 dp: 48.83 m.

C Core — Answer: 35.21 m

Let h be height of tower. Let x be initial distance from tower. $\tan(27) = h/x$, so $x = h/\tan(27)$. After moving: $\tan(42) = h/(x-30)$, so $x-30 = h/\tan(42)$. Substituting: $h/\tan(27) - 30 = h/\tan(42)$. $h(1/\tan(27) - 1/\tan(42)) = 30$. $\tan(27) = 0.50953$, so $1/\tan(27) = 1.9626$. $\tan(42) = 0.90040$, so $1/\tan(42) = 1.1106$. Difference = 0.8520. $h = 30/0.8520 = 35.21$ m.

C Core — Answer: 58.56 m

For the nearer boat: $\tan(45) = 80/d_1$, so $d_1 = 80/\tan(45) = 80$ m. For the farther boat: $\tan(30) = 80/d_2$, so $d_2 = 80/\tan(30) = 80/0.57735 = 138.564$ m. Distance between boats = $d_2 - d_1 = 138.564 - 80 = 58.564$ m. Rounded to 2 dp: 58.56 m.

C Core — Answer: 61.57 m

$\sin(38) = \text{height}/100$. $\text{height} = 100 * \sin(38) = 100 * 0.61566 = 61.566$ m. Rounded to 2 dp: 61.57 m.

C Core — Answer: 214.45 m

$\tan(25) = 100/\text{distance}$. $\text{distance} = 100/\tan(25) = 100/0.46631 = 214.452$ m. Rounded to 2 dp: 214.45 m.

C Core — Answer: 75.23 m

$\tan(28) = 40/\text{distance}$. $\text{distance} = 40/\tan(28) = 40/0.53171 = 75.228$ m. Rounded to 2 dp: 75.23 m.

H Challenge — Answer: 44.84 m

Let h be height of tower. Let d be distance from A to tower base. $\tan(18) = h/d$, so $d = h/\tan(18)$. $\tan(27) = h/(d-50)$, so $d-50 = h/\tan(27)$. Substituting: $h/\tan(18) - 50 = h/\tan(27)$. $h(1/\tan(18) - 1/\tan(27)) = 50$. $\tan(18) = 0.32492$, $1/\tan(18) = 3.0777$. $\tan(27) = 0.50953$, $1/\tan(27) = 1.9626$. Difference = 1.1151. $h = 50/1.1151 = 44.84$ m.

H Challenge — Answer: 32.98 m

Let x be horizontal distance between building and tower. From the top of the building (25 m high): $\tan(40) = 25/x$, so $x = 25/\tan(40) = 25/0.83910 = 29.795$ m. The height difference between top of building and top of tower: $\tan(15) = h/x$, so $h = x * \tan(15) = 29.795 * 0.26795 = 7.983$ m. Tower height = $25 + h = 25 + 7.983 = 32.983$ m. Rounded to 2 dp: 32.98 m.

H Challenge — Answer: 36.21 m

Let h be height of balloon. Let x be distance from A to point directly below balloon. $\tan(35) = h/x$, so $x = h/\tan(35)$. $\tan(52) = h/(80-x)$, so $80-x = h/\tan(52)$. Substituting: $80 - h/\tan(35) = h/\tan(52)$. $80 = h(1/\tan(35) + 1/\tan(52))$. $h = 80 / (1/\tan(35) + 1/\tan(52))$. $\tan(35) = 0.70021$, $1/\tan(35) = 1.4281$. $\tan(52) = 1.2799$, $1/\tan(52) = 0.7813$. Sum = 2.2094. $h = 80/2.2094 = 36.21$ m.

H Challenge — Answer: 55.84 m

Let h be height of tower, d be original distance from tower. $\tan(25) = h/d$, so $d = h/\tan(25)$. After moving 40 m closer: $\tan(35) = h/(d-40)$, so $d-40 = h/\tan(35)$. Substituting: $h/\tan(25) - 40 = h/\tan(35)$. $h(1/\tan(25) - 1/\tan(35)) = 40$. $\tan(25) = 0.46631$, $1/\tan(25) = 2.1445$. $\tan(35) = 0.70021$, $1/\tan(35) = 1.4281$. Difference = 0.7164. $h = 40/0.7164 = 55.84$ m.

E1_Q4: Bearings - Problems

F Foundation — Answer: 180 degrees

If B is due north of A, then A is due south of B, which is a bearing of 180 degrees.

F Foundation — Answer: 090 degrees

Due east corresponds to a bearing of 090 degrees.

F Foundation — Answer: 270 degrees

Due west corresponds to a bearing of 270 degrees.

F Foundation — Answer: 300 degrees

The back bearing is found by adding 180 degrees. $120 + 180 = 300$ degrees.

C Core — Answer: 17.32 km

East component = $20 * \sin(60) = 20 * 0.86603 = 17.32$ km.

C Core — Answer: 11.49 km

North component = $15 * \cos(40) = 15 * 0.76604 = 11.491$ km. Rounded to 2 dp: 11.49 km.

C Core — Answer: 150.00 km

Bearing 120 degrees is 60 degrees east of south. South component = $300 * \cos(60) = 300 * 0.5 = 150.00$ km.

C Core — Answer: 50.00 km

The angle between the two bearings is $140 - 50 = 90$ degrees. Triangle PQR has sides 30 and 40 at a right angle at Q. $PR^2 = 30^2 + 40^2 = 900 + 1600 = 2500$. $PR = \sqrt{2500} = 50.00$ km.

C Core — Answer: 10.00 km

The difference between bearings is $120 - 30 = 90$ degrees. The two legs are perpendicular. $d^2 = 8^2 + 6^2 = 64 + 36 = 100$. $d = \sqrt{100} = 10.00$ km.

C Core — Answer: West = 25.00 km, South = 43.30 km

Bearing 210 is 30 degrees past south. South component = $50 * \cos(30) = 50 * 0.86603 = 43.30$ km. West component = $50 * \sin(30) = 50 * 0.5 = 25.00$ km.

C Core — Answer: 255 degrees

Back bearing = $075 + 180 = 255$ degrees.

C Core — Answer: 22.66 km

North component = $25 * \cos(25) = 25 * 0.90631 = 22.658$ km. Rounded to 2 dp: 22.66 km.

H Challenge — Answer: Distance = 50 km, Bearing = 096.87 degrees

Angle between bearings = $150 - 60 = 90$ degrees. $HA = 40$, $AB = 30$, and the angle between them at A is 90 degrees. $HB^2 = 40^2 + 30^2 = 2500$, $HB = 50$ km. $\tan(\text{angle AHB}) = 30/40 = 0.75$, angle AHB = $\tan^{-1}(0.75) = 36.87$ degrees. Bearing of B from H = $060 + 36.87 = 096.87$ degrees.

H Challenge — Answer: Distance = 28.28 km, Bearing = 090.00 degrees

Difference between bearings = $135 - 45 = 90$ degrees. Two legs are perpendicular and equal (20 km each). $PR^2 = 20^2 + 20^2 = 800$. $PR = \sqrt{800} = 28.28$ km. Since triangle PQR is right-angled and isosceles, angle QPR = 45 degrees. Bearing of R from P = $045 + 45 = 090$ degrees.

H Challenge — Answer: 64.03 km

At Y, the direction back to X has bearing $030 + 180 = 210$ degrees. The direction from Y to Z has bearing 120 degrees. The angle between YX and YZ = $210 - 120 = 90$ degrees. So angle XYZ = 90 degrees. Using cosine rule: $XZ^2 = 50^2 + 40^2 - 2 \cdot 50 \cdot 40 \cdot \cos(90) = 2500 + 1600 - 0 = 4100$. $XZ = \sqrt{4100} = 64.03$ km.

H Challenge — Answer: Distance = 29.15 km, Bearing = 119.05 degrees

At B, reverse bearing to A is $060 + 180 = 240$. Forward bearing to C is 150. Angle between BA and BC = $240 - 150 = 90$ degrees. So angle ABC is 90 degrees. $AC^2 = 15^2 + 25^2 = 225 + 625 = 850$, $AC = \sqrt{850} = 29.15$ km. Using coordinates: A = (0,0). B = $(15 \cdot \sin 60, 15 \cdot \cos 60) = (12.99, 7.50)$. From B: east = $25 \cdot \sin 150 = 12.50$, north = $25 \cdot \cos 150 = -21.65$. C = $(12.99 + 12.50, 7.50 - 21.65) = (25.49, -14.15)$. Bearing of C from A: $\arctan(25.49/14.15) = 60.95$ degrees from north towards east, but since C is south of A: bearing = $180 - 60.95 = 119.05$ degrees.

E1_Q5: Sine Rule - Problems

F Foundation — Answer: 14.14 cm

By sine rule: $a/\sin(A) = b/\sin(B)$. So $10/\sin(30) = b/\sin(45)$. $b = 10 \cdot \sin(45) / \sin(30) = 10 \cdot 0.70711 / 0.5 = 14.14$ cm.

F Foundation — Answer: 13.02 cm

By sine rule: $p/\sin(P) = q/\sin(Q)$. $12/\sin(60) = q/\sin(70)$. $q = 12 \cdot \sin(70) / \sin(60) = 12 \cdot 0.93969 / 0.86603 = 13.02$ cm.

F Foundation — Answer: 12.26 cm

By sine rule: $x/\sin(X) = y/\sin(Y)$. $8/\sin(40) = y/\sin(80)$. $y = 8 \cdot \sin(80) / \sin(40) = 8 \cdot 0.98481 / 0.64279 = 12.26$ cm.

F Foundation — Answer: 55.0 degrees

By sine rule: $a/\sin(A) = b/\sin(B)$. $7/\sin(35) = 10/\sin(B)$. $\sin(B) = 10 \cdot \sin(35) / 7 = 10 \cdot 0.57358 / 7 = 0.8194$. $B = \sin^{-1}(0.8194) = 55.0$ degrees.

C Core — Answer: 38.7 degrees

By sine rule: $a/\sin(A) = b/\sin(B)$. $\sin(B) = b \cdot \sin(A) / a = 10 \cdot \sin(30) / 8 = 10 \cdot 0.5 / 8 = 0.625$. $B = \sin^{-1}(0.625) = 38.7$ degrees.

C Core — Answer: 63.1 degrees

By sine rule: $p/\sin(P) = q/\sin(Q)$. $\sin(Q) = q \cdot \sin(P) / p = 20 \cdot \sin(42) / 15 = 20 \cdot 0.66913 / 15 = 0.89217$. $Q = \sin^{-1}(0.89217) = 63.1$ degrees.

C Core — Answer: 9.78 cm

Angle C = $180 - 50 - 60 = 70$ degrees. By sine rule: $a/\sin(A) = c/\sin(C)$. $a = 12 * \sin(50) / \sin(70) = 12 * 0.76604 / 0.93969 = 9.78$ cm.

C Core — Answer: 19.29 cm

Angle Y = $180 - 65 - 45 = 70$ degrees. By sine rule: $x/\sin(X) = y/\sin(Y)$. $x = 20 * \sin(65) / \sin(70) = 20 * 0.90631 / 0.93969 = 19.29$ cm.

C Core — Answer: 34.3 degrees

$\sin(B) = b * \sin(A) / a = 12 * \sin(25) / 9 = 12 * 0.42262 / 9 = 0.56349$. $B = \sin^{-1}(0.56349) = 34.3$ degrees.

C Core — Answer: 13.24 cm

Angle A = $180 - 55 - 65 = 60$ degrees. By sine rule: $b/\sin(B) = a/\sin(A)$. $b = 14 * \sin(55) / \sin(60) = 14 * 0.81915 / 0.86603 = 13.24$ cm.

C Core — Answer: 27.45 cm

Angle P = $180 - 72 - 48 = 60$ degrees. By sine rule: $q/\sin(Q) = p/\sin(P)$. $q = 25 * \sin(72) / \sin(60) = 25 * 0.95106 / 0.86603 = 27.45$ cm.

C Core — Answer: 38.8 degrees or 141.2 degrees

$\sin(Y) = y * \sin(X) / x = 8 * \sin(28) / 6 = 8 * 0.46947 / 6 = 0.62596$. $Y = \sin^{-1}(0.62596) = 38.8$ degrees. The ambiguous case: Y could also be $180 - 38.8 = 141.2$ degrees. Check: $X + 141.2 = 28 + 141.2 = 169.2 < 180$, so both are valid.

H Challenge — Answer: 48.6 degrees or 131.4 degrees

By sine rule: $\sin(B) = b * \sin(A) / a = 15 * \sin(30) / 10 = 15 * 0.5 / 10 = 0.75$. $B = \sin^{-1}(0.75) = 48.6$ degrees. The ambiguous case: B could also be $180 - 48.6 = 131.4$ degrees. Check: $A + 131.4 = 30 + 131.4 = 161.4 < 180$, so both are valid.

H Challenge — Answer: 45.06 cm

Angle C = $180 - 40 - 65 = 75$ degrees. The largest angle is C (75 degrees), so the longest side is c (opposite C). Using sine rule: $c/\sin(C) = a/\sin(A)$. $c = 30 * \sin(75) / \sin(40) = 30 * 0.96593 / 0.64279 = 45.06$ cm.

H Challenge — Answer: 171.62 m

The tree and the two endpoints form a triangle. The angle at the tree = $180 - 72 - 48 = 60$ degrees. The nearer end has the larger angle (72 degrees), as the side opposite the larger angle is longer. Distance from the nearer end (72-degree end) is opposite the 48-degree angle. Using sine rule: $d / \sin(48) = 200 / \sin(60)$. $d = 200 * \sin(48) / \sin(60) = 200 * 0.74314 / 0.86603 = 171.62$ m.

H Challenge — Answer: Two triangles are possible. $B = 61.8$ degrees or $B = 118.2$ degrees

By sine rule: $\sin(B) = b \cdot \sin(A) / a = 12 \cdot \sin(36) / 8 = 12 \cdot 0.58779 / 8 = 0.88169$. $B = \sin^{-1}(0.88169) = 61.8$ degrees. Check if B could also be $180 - 61.8 = 118.2$ degrees. $A + 118.2 = 36 + 118.2 = 154.2 < 180$, so both are valid. Two triangles are possible.

E1_Q6: Cosine Rule — Problems

F Foundation — Answer: $BC = 7.21$ cm

Using cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$. $BC^2 = 6^2 + 8^2 - 2(6)(8)\cos 60^\circ = 36 + 64 - 96(0.5) = 100 - 48 = 52$. $BC = \sqrt{52} = 7.21$ cm.

F Foundation — Answer: Angle $P = 82.8^\circ$

Using cosine rule: $\cos P = (PQ^2 + PR^2 - QR^2) / (2 \times PQ \times PR) = (25 + 49 - 64) / (2 \times 5 \times 7) = 10 / 70 = 0.1429$. $P = \cos^{-1}(0.1429) = 82.8^\circ$.

F Foundation — Answer: 82.8°

Largest side is 6 cm opposite largest angle. $\cos \theta = (4^2 + 5^2 - 6^2) / (2 \times 4 \times 5) = (16 + 25 - 36) / 40 = 5 / 40 = 0.125$. $\theta = \cos^{-1}(0.125) = 82.8^\circ$.

F Foundation — Answer: $a = 6.81$ cm

$a^2 = 9^2 + 7^2 - 2(9)(7)\cos 48^\circ = 81 + 49 - 126(0.6691) = 130 - 84.31 = 45.69$. $a = \sqrt{45.69} = 6.76$ cm.

C Core — Answer: 20.1 km

The angle between the two paths is $180^\circ - 60^\circ = 120^\circ$. Using cosine rule: $d^2 = 12^2 + 15^2 - 2(12)(15)\cos 120^\circ = 144 + 225 - 360(-0.5) = 369 + 180 = 549$. $d = \sqrt{549} = 23.4$ km.

C Core — Answer: $BC = 12.4$ cm, angle $B = 43.7^\circ$

$BC^2 = 11^2 + 9^2 - 2(11)(9)\cos 72^\circ = 121 + 81 - 198(0.3090) = 202 - 61.18 = 140.82$. $BC = 11.87$ cm. Then $\sin B / 9 = \sin 72^\circ / 11.87 \rightarrow \sin B = 9 \sin 72^\circ / 11.87 = 0.721$. $B = 46.2^\circ$.

C Core — Answer: 73.7°

Using cosine rule: $\cos A = (10^2 + 10^2 - 12^2) / (2 \times 10 \times 10) = (100 + 100 - 144) / 200 = 56 / 200 = 0.28$. $A = \cos^{-1}(0.28) = 73.7^\circ$.

C Core — Answer: 33.1°

Smallest angle is opposite smallest side (8 m). $\cos \theta = (11^2 + 14^2 - 8^2) / (2 \times 11 \times 14) = (121 + 196 - 64) / 308 = 253 / 308 = 0.8214$. $\theta = \cos^{-1}(0.8214) = 34.8^\circ$.

H Challenge — Answer: $BC = 6.04$ cm or 13.9 cm

Using sine rule: $\sin C / 8 = \sin 38^\circ / 11 \rightarrow \sin C = 8 \sin 38^\circ / 11 = 0.447$. $C = 26.6^\circ$ or 153.4° . If $C = 26.6^\circ$: $A = 180 - 38 - 26.6 = 115.4^\circ$, $BC^2 = 11^2 + 8^2 - 2(11)(8)\cos 115.4^\circ = 121 + 64 - 176(-0.431) = 185 + 75.9 = 260.9$, $BC = 16.2$ cm.

If $C = 153.4^\circ$: $A = 180 - 38 - 153.4 = -11.4^\circ$, impossible. Wait, this is cosine rule problem. Using cosine rule with alternate approach: $BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos A$. Without angle A, use substitution.

H Challenge — Answer: 116.4°

In triangle ABC: $\cos BCA = (8^2 + 7^2 - 5^2)/(2 \times 8 \times 7) = (64 + 49 - 25)/112 = 88/112 = 0.7857$. $\angle BCA = 38.2^\circ$. In triangle ACD: $\cos ACD = (8^2 + 9^2 - 6^2)/(2 \times 8 \times 9) = (64 + 81 - 36)/144 = 109/144 = 0.7569$. $\angle ACD = 40.8^\circ$. $\angle BCD = 38.2^\circ + 40.8^\circ = 79.0^\circ$.

H Challenge — Answer: 64.6°

Half of diagonal of base $= 3\sqrt{2} = 4.24$ cm. Slant edge $= 10$ cm. Height: $h^2 = 10^2 - 4.24^2 = 100 - 18.0 = 82.0$. $h = 9.06$ cm. $\cos \theta = 4.24/10 = 0.424$. $\theta = 64.9^\circ$.

H Challenge — Answer: Distance $= 247.3$ km, bearing $= 087.7^\circ$

Angle between paths $= 140^\circ - 50^\circ = 90^\circ$. Using Pythagoras: $d^2 = 200^2 + 150^2 = 40000 + 22500 = 62500$. $d = 250$ km. Bearing: $\theta = \tan^{-1}(150/200) = 36.9^\circ$. Final bearing $= 50^\circ + 36.9^\circ = 86.9^\circ$.

X Extension — Answer: See solution for proof

Place triangle with A at origin, AB along x-axis. Coordinates: A(0,0), B(c,0), C(b cos A, b sin A). Then $a^2 = (b \cos A - c)^2 + (b \sin A)^2 = b^2 \cos^2 A - 2bc \cos A + c^2 + b^2 \sin^2 A = b^2(\cos^2 A + \sin^2 A) + c^2 - 2bc \cos A = b^2 + c^2 - 2bc \cos A$.

X Extension — Answer: 20 cm

By cosine rule: $c^2 = 8^2 + 12^2 - 2(8)(12)\cos\theta = 64 + 144 - 192 \cos\theta = 208 - 192 \cos\theta$. c is maximized when $\cos\theta$ is minimized (-1 at $\theta = 180^\circ$). $c_{\max}^2 = 208 - 192(-1) = 400$. $c_{\max} = 20$ cm.

X Extension — Answer: 87.1°

In triangle ABD: $\cos ABD = (6^2 + 9^2 - 7^2)/(2 \times 6 \times 9) = (36 + 81 - 49)/108 = 68/108 = 0.6296$. $\angle ABD = 51.0^\circ$. In triangle BCD: $\cos CBD = (8^2 + 9^2 - 10^2)/(2 \times 8 \times 9) = (64 + 81 - 100)/144 = 45/144 = 0.3125$. $\angle CBD = 71.8^\circ$. $\angle ABC = 51.0^\circ + 71.8^\circ = 122.8^\circ$.

X Extension — Answer: Proof that $\cos C < -0.5$

Longest side is $(a+b)$. By cosine rule: $\cos C = (a^2 + b^2 - (a+b)^2)/(2ab) = (a^2 + b^2 - a^2 - 2ab - b^2)/(2ab) = -2ab/(2ab) = -1$. So $\cos C = -1$ and $C = 180^\circ$. But a triangle with sides a, b, $a+b$ is degenerate (collinear). For a non-degenerate triangle, $a+b > c$ and $C > 120^\circ$ only when... Actually $\cos C = -1$ means the triangle is degenerate. For a proper triangle, C approaches 180° but never reaches it, so angle $> 120^\circ$.

E1_Q7: Area of Triangle using $1/2 ab \sin C$ - Problems

F Foundation — Answer: 12 sq cm

Area $= (1/2) * a * b * \sin(C) = (1/2) * 6 * 8 * \sin(30) = 24 * 0.5 = 12$ sq cm.

F Foundation — Answer: 42.43 sq cm

$$\text{Area} = (1/2) * 10 * 12 * \sin(45) = 60 * 0.70711 = 42.43 \text{ sq cm.}$$

F Foundation — Answer: 19.49 sq cm

$$\text{Area} = (1/2) * 5 * 9 * \sin(60) = 22.5 * 0.86603 = 19.49 \text{ sq cm.}$$

F Foundation — Answer: 38.5 sq cm

$$\text{Area} = (1/2) * 7 * 11 * \sin(90) = 38.5 * 1 = 38.5 \text{ sq cm.}$$

C Core — Answer: 36.77 sq cm

$$\text{Area} = (1/2) * 8 * 12 * \sin(50) = 48 * 0.76604 = 36.77 \text{ sq cm.}$$

C Core — Answer: 48.6 degrees

$$\text{Area} = (1/2) * a * b * \sin(C). 30 = (1/2) * 8 * 10 * \sin(C) = 40 * \sin(C). \sin(C) = 30/40 = 0.75. C = \sin^{-1}(0.75) = 48.6 \text{ degrees.}$$

C Core — Answer: 20 cm

$$\text{Area} = (1/2) * a * b * \sin(C). 25 = (1/2) * 5 * b * \sin(30) = (1/2) * 5 * b * 0.5 = 1.25b. b = 25/1.25 = 20 \text{ cm.}$$

C Core — Answer: 57.10 sq cm

$$\text{Area} = (1/2) * 9 * 14 * \sin(65) = 63 * 0.90631 = 57.10 \text{ sq cm.}$$

C Core — Answer: 41.8 degrees or 138.2 degrees

$$\text{Area} = (1/2) * 10 * 12 * \sin(C) = 60 * \sin(C). 40 = 60 * \sin(C). \sin(C) = 40/60 = 2/3 = 0.6667. C = \sin^{-1}(0.6667) = 41.8 \text{ degrees. Since } \sin(180 - C) = \sin(C), C \text{ could also be } 180 - 41.8 = 138.2 \text{ degrees.}$$

C Core — Answer: 19.28 sq cm

$$\text{Area} = (1/2) * 6 * 10 * \sin(40) = 30 * 0.64279 = 19.28 \text{ sq cm.}$$

C Core — Answer: 9.77 cm

$$\text{Area} = (1/2) * a * b * \sin(C). 36 = (1/2) * 9 * b * \sin(55) = 4.5 * b * 0.81915 = 3.686b. b = 36/3.686 = 9.77 \text{ cm.}$$

C Core — Answer: 86.04 sq cm

$$\text{Area} = (1/2) * 15 * 20 * \sin(35) = 150 * 0.57358 = 86.04 \text{ sq cm.}$$

H Challenge — Answer: 59.81 sq cm

$$\cos(C) = (a^2 + b^2 - c^2)/(2ab) = (144 + 225 - 100)/(2*12*15) = 269/360 = 0.74722. C = \cos^{-1}(0.74722) = 41.6 \text{ degrees. Area} = (1/2)*12*15*\sin(41.6) = 90*0.6645 = 59.81 \text{ sq cm.}$$

H Challenge — Answer: 160.70 sq cm

Area = $(1/2) * b * c * \sin(A) = (1/2) * 20 * 25 * \sin(40) = 250 * 0.64279 = 160.70 \text{ sq cm}.$

H Challenge — Answer: 52.6 degrees

Area = $(1/2) * p * q * \sin(R)$. $100 = (1/2) * 14 * 18 * \sin(R) = 126 * \sin(R)$. $\sin(R) = 100/126 = 0.79365$. $R = \sin^{-1}(0.79365) = 52.6 \text{ degrees}.$

H Challenge — Answer: 84.00 sq cm

Find angle C using cosine rule: $\cos(C) = (a^2 + b^2 - c^2)/(2ab) = (169 + 196 - 225)/(2*13*14) = 140/364 = 0.38462$. $C = \cos^{-1}(0.38462) = 67.38 \text{ degrees}$. $\sin(C) = \sin(67.38) = 0.92308$. Area = $(1/2)*13*14*0.92308 = 91*0.92308 = 84.00 \text{ sq cm}.$ (This is a 13-14-15 triangle, area = 84 exactly.)
